



Requirements

- **BMETE12AP50** (Physicist-Engineer BSc, **4** credits) with exercises every 2nd week.
- **BMETEAFBsFELEK-00** (Physics BSc, English, **3** credits) no exercises.
- **Moodle access at Electronics - BMETE12AP50**



Electronics

Topics

Week	Lecture	Topics	Book chapter (Eggleston)
1 (E2-1)	18-Feb	Basic concepts and resistor (DC) circuits	Chapter-1
2 (E1-1)	25-Feb	DC, Operational amplifiers (1) – DC and linear	Chapters-1, 6
3 (E2-2)	04-Mar	AC circuits (1) – components and RC circuits	Chapters-1, 2
4 (E1-2)	11-Mar	AC (2) – RLC, transformers	Chapter-2
5 (E2-3)	18-Mar	Diode circuits and power management	Chapter-3
6 (E1-3)	25-Mar	BJTs (1) – DC and switching applications	Chapter-4
7 (E2-1)	01-Apr	Bipolar junction transistors (2) – amplifiers	Chapter-4
8	08-Apr	Spring break	
9 (E1-4)	15-Apr	Dean's break	
10 (E2-4)	22-Apr	Op. amps. (2) – AC and nonlinear applications	Chapter-6
11 (E1-5)	29-Apr	FETs (1) – DC and switching applications	Chapter-5
12 (E2-5)	06-May	Indoor Test	
13 (E1-6)	13-May	Field-effect transistors (2) – amplifiers	Chapter-5
14 (E2-6)	20-May	Oscillators	Chapter-7
15 (E1-7)	27-May	Digital circuits. Mixed signal (dig. + analogue)	Chapter-8



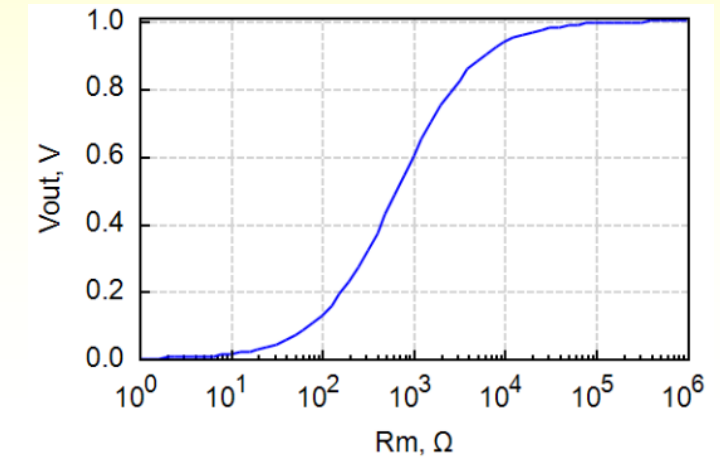
Tools (optional)

1. SMath Studio:

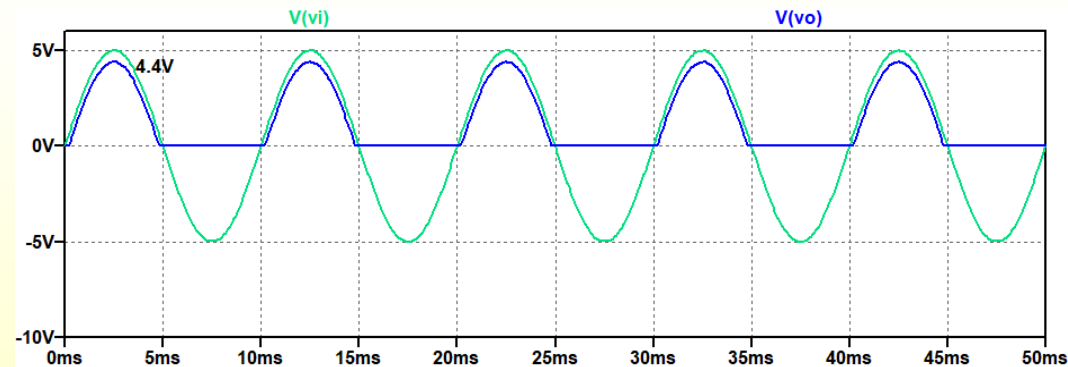
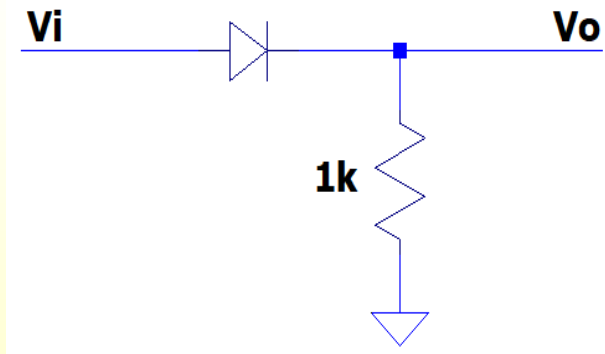
$$R_1 := 1 \text{ k}\Omega \quad R_2 := 2 \text{ k}\Omega \quad V_{in} := 3 \text{ V}$$

$$V_{out}(R_m) := \frac{V_{in}}{R_2} \cdot \left(\frac{1}{R_2} + \frac{1}{R_1} + \frac{1}{R_m} \right)^{-1}$$

$$V_{out}(100 \text{ }\Omega) = 0.13 \text{ V} \quad V_{out}(1 \text{ k}\Omega) = 0.6 \text{ V} \quad V_{out}(50 \text{ k}\Omega) = 0.987 \text{ V}$$

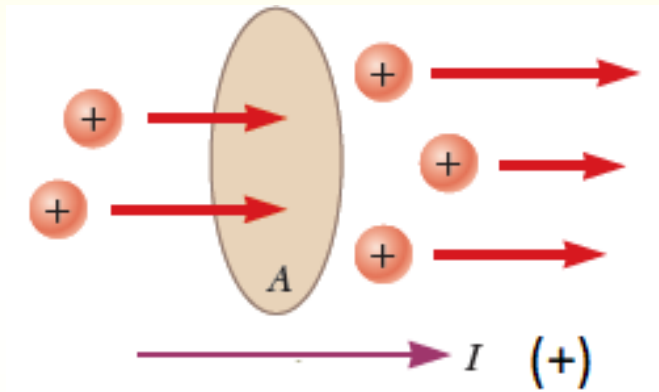


2. LTspice: (www.analog.com)

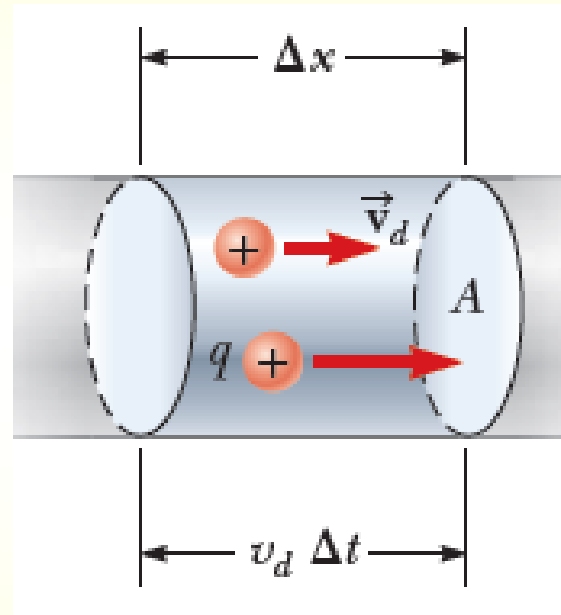


Physics background

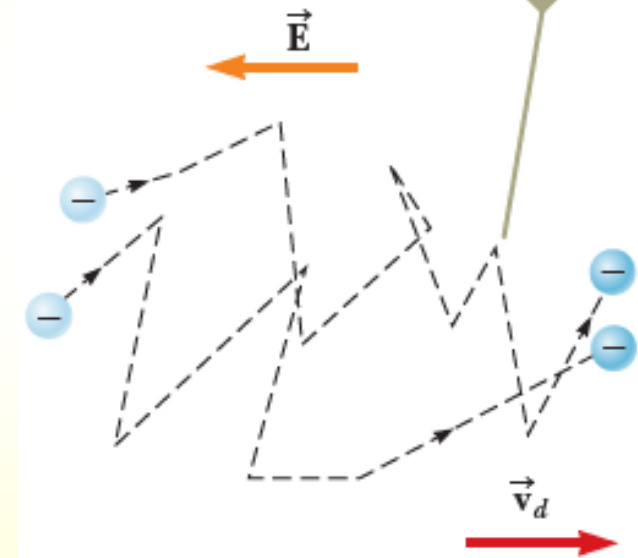
- Current



$$I_{\text{avg}} = \frac{\Delta Q}{\Delta t} = \frac{(nAv_d \Delta t)q}{\Delta t}$$



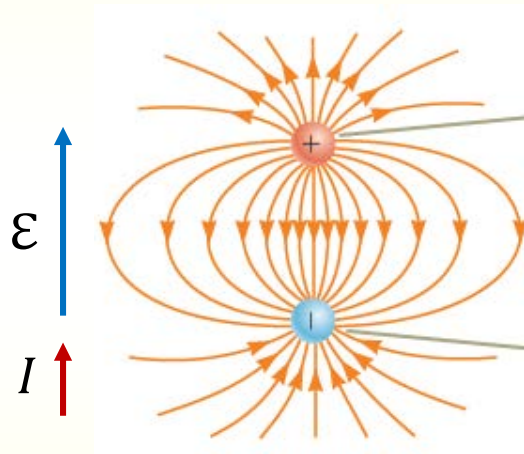
The random motion of the charge carriers is modified by the field, and they have a drift velocity opposite the direction of the electric field.





Physics background

- Voltage & potential difference



$$U_{AB} = V_{\textcircled{B}} - V_{\textcircled{A}} = \Delta V = - \int_{\textcircled{A}}^{\textcircled{B}} \vec{\mathbf{E}} \cdot d\vec{\mathbf{s}} = - \int_{\textcircled{A}}^{\textcircled{B}} E ds$$

$\Delta V = -Ed$

U_{field}

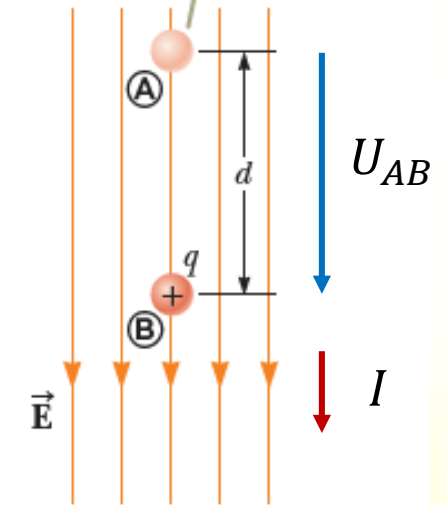
$U_{\text{field}} \sim$ work done by field

U_{AB} : voltage (drop)

$\mathcal{E}$: electromotive force of battery

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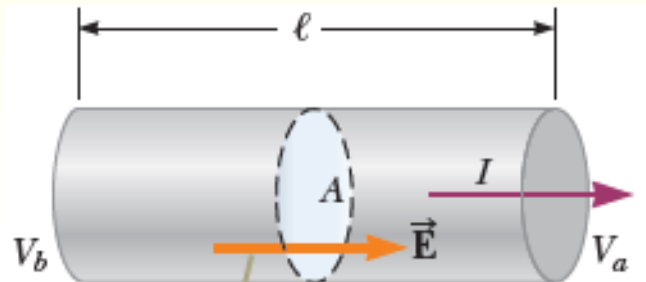
When a positive charge moves from point $\textcircled{A}$ to point $\textcircled{B}$, the electric potential energy of the charge-field system decreases.





Physics background

- Ohm's law

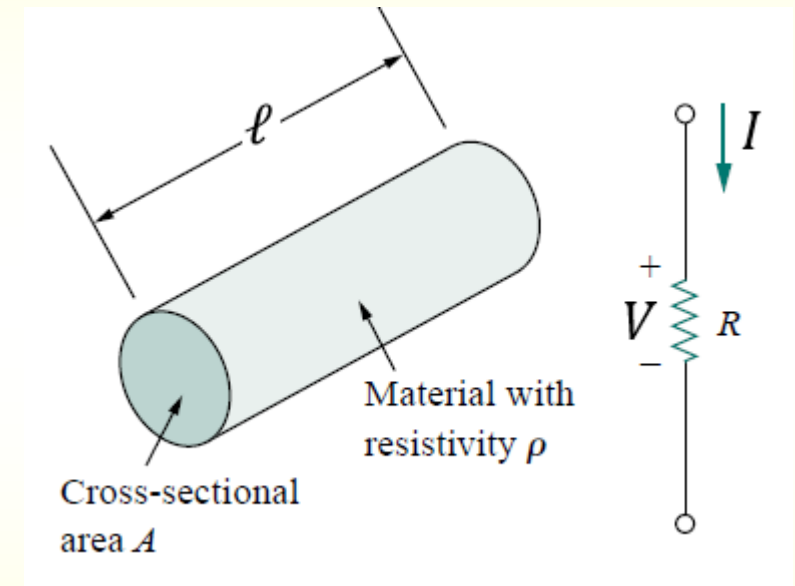


A potential difference $\Delta V = V_b - V_a$ maintained across the conductor sets up an electric field $\vec{E}$, and this field produces a current I that is proportional to the potential difference.

$$J = \frac{I}{A} = nv_d q$$

$$J = \sigma E = \sigma \frac{\Delta V}{\ell}$$

$$U = \Delta V = \frac{J\ell}{\sigma} = \frac{\ell}{\sigma A} I = \varrho \frac{\ell}{A} I = RI$$





Physics background

- Ohm's law and other definitions

Resistance

$$R = \frac{V}{I} = \varrho \frac{\ell}{A} [\Omega]$$

Resistivity
(specific resistance)

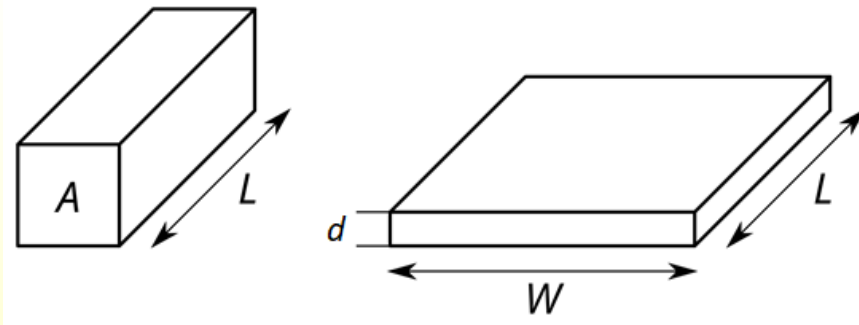
$$\varrho [\Omega\text{m}]$$

Sheet resistance
 $R_s|_{(L=W)} [\Omega]$

$$R = \frac{\varrho}{d} \frac{L}{W} = R_s \frac{L}{W}$$

Resistance per unit length
(e.g. of a long wire)

$$\frac{R}{L} \left[\frac{\Omega}{\text{m}} \right]$$





Physics background

Electric field in a circuit:

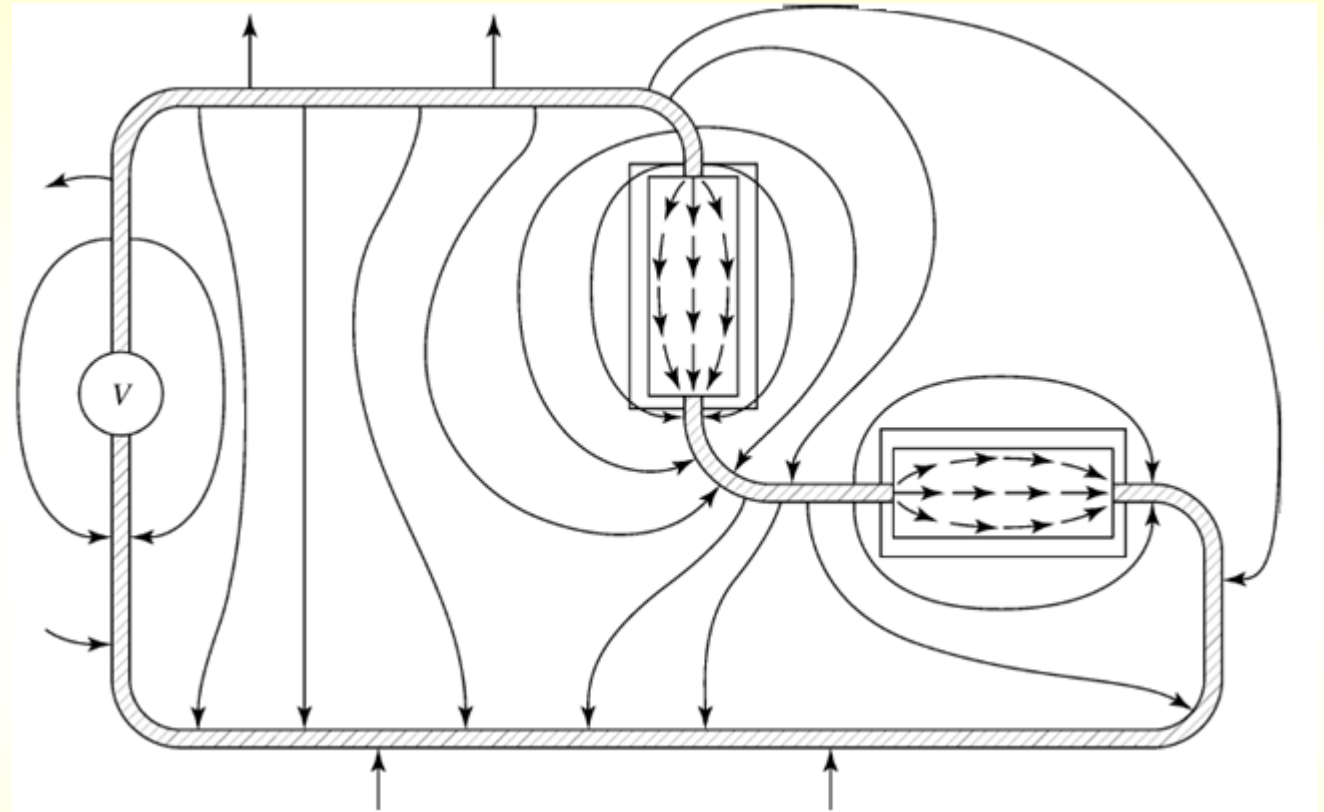
- Wires are not perfectly equipotential
→ $\underline{E}$ is not perfectly orthogonal

Validity of Kirchhoff's loop law (KVL):

- In a DC (and quasi-static AC) circuits, the electric field is conservative.

Approximation:

- There is no significantly changing magnetic field linking the closed loop
($\oint \underline{E} d\mathbf{r} = -\frac{d\Phi_B}{dt} \approx 0$)
- The size of the circuit is much smaller than the wavelength of the electrical signals (low-frequency limit).





Resistors

- Colour chart & standard values

E3	20%	10	22	47						
E6	10%	10	15	22	33	47	68			
E12	5%	10	12	15	18	22	27	33	39	47
		56	68	82						
E24	2%	10	11	12	13	15	16	18	20	22
		24	27	30	33	36	39	43	47	51
		56	62	68	75	82	91			
E96	1%	100	102	105	107	110	113	115	118	121
		124	127	130	133	137	140	143	147	150
		154	158	162	165	169	174	178	182	187
		191	196	200	205	210	215	221	226	232
		237	243	249	255	261	267	274	280	287
		294	301	309	316	324	332	340	348	357
		365	374	383	392	402	412	422	432	442
		453	464	475	487	499	511	523	536	549
		562	576	590	604	619	634	649	665	681
		698	715	732	750	768	787	806	825	845
		866	887	909	931	953	976			

www.resistorguide.com

	Color	Significant figures			Multiply	Tolerance (%)	Temp. Coeff. (ppm/K)	Fail Rate (%)
Bad	black	0	0	0	x 1		250 (U)	
Beer	brown	1	1	1	x 10	1 (F)	100 (S)	1
Rots	red	2	2	2	x 100	2 (G)	50 (R)	0.1
Our	orange	3	3	3	x 1K		15 (P)	0.01
Young	yellow	4	4	4	x 10K		25 (Q)	0.001
Guts	green	5	5	5	x 100K	0.5 (D)	20 (Z)	
But	blue	6	6	6	x 1M	0.25 (C)	10 (Z)	
Vodka	violet	7	7	7	x 10M	0.1 (B)	5 (M)	
Goes	grey	8	8	8	x 100M	0.05 (A)	1(K)	
Well	white	9	9	9	x 1G			
Get	gold			3th digit only for 5 and 6 bands	x 0.1	5 (J)		
Some	silver				x 0.01	10 (K)		
Now!	none					20 (M)		

6 band 3.21kΩ 1% 50ppm/K

5 band 521Ω 1%

4 band 82kΩ 5%

3 band 330Ω 20%

gap between band 3 and 4 indicates reading direction



Resistors

- Prefixes & resistivity

Multiple	Prefix	Symbol
10^{12}	tera	T
10^9	giga	G
10^6	mega	M
10^3	kilo	k

Multiple	Prefix	Symbol
10^{-3}	milli	m
10^{-6}	micro	μ
10^{-9}	nano	n
10^{-12}	pico	p
10^{-15}	femto	f

Material	Resistivity ($\Omega \cdot m$)	Usage
Silver	1.64×10^{-8}	Conductor
Copper	1.72×10^{-8}	Conductor
Aluminum	2.8×10^{-8}	Conductor
Gold	2.45×10^{-8}	Conductor
Carbon	4×10^{-5}	Semiconductor
Germanium	47×10^{-2}	Semiconductor
Silicon	6.4×10^2	Semiconductor
Paper	10^{10}	Insulator
Mica	5×10^{11}	Insulator
Glass	10^{12}	Insulator
Teflon	3×10^{12}	Insulator



Resistors

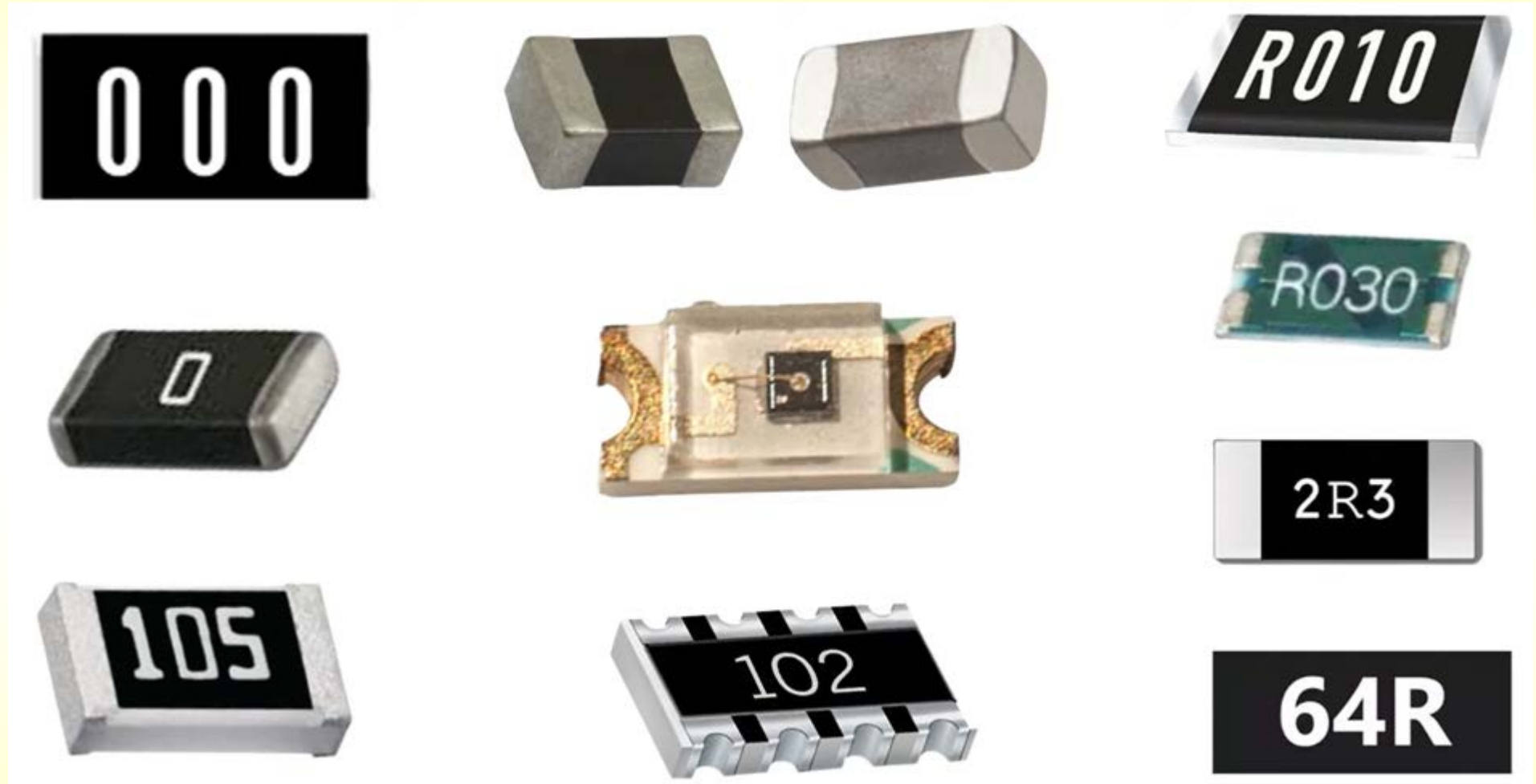
- Types





Resistors

- Types



Resistors

- Types

Digital potentiometer ↓

